

WEEK - 3 PRACTICAL

2520090198 Siri Chandana

The screenshot shows the MongoDB Data Explorer interface. On the left, the 'CLUSTERS (1)' sidebar lists 'Cluster0' with sub-clusters 'admin', 'bookflow', 'book_metadata', 'local', and 'studentDB'. The 'book_metadata' cluster is selected. The main panel displays 'Cluster0 > bookflow > book_metadata' with tabs for 'Documents', 'Aggregations', 'Schema', 'Indexes', and 'Validation'. The 'Documents' tab is active, showing a single document with the following fields: `_id` (ObjectId), `book_id` (101), `title` ("The Martian"), `author` ("Andy Weir"), `published_year` (2021), `category` ("Science Fiction"), `format` ("Digital PDF"), `file_size` ("2.5 MB"), `member_id` (101), and `reviews` (Array (3)). The bottom status bar indicates 'Alert System Status: All Good' and '©2026 MongoDB, Inc.'.

The screenshot shows the MongoDB Data Explorer interface with the 'book_metadata' cluster selected. The main panel displays 'Cluster0 > bookflow > book_metadata' with tabs for 'Documents', 'Aggregations', 'Schema', 'Indexes', and 'Validation'. The 'Documents' tab is active, showing a list of documents. The first document is expanded, showing the following fields: `_id` (ObjectId), `book_id` (105), `title` ("Data Structures"), `author` ("Mark Allen"), `published_year` (2024), `category` ("Computer Science"), `format` ("Digital PDF"), `file_size` ("5.0 MB"), `member_id` (105), and `reviews` (Array (2)). The bottom status bar indicates 'Alert System Status: All Good' and '©2026 MongoDB, Inc.'.

Cluster0

book_metadata

Cluster0 > bookflow > book_metadata

View monitoringVisualize Your Data

Documents5

Aggregations

Schema

Indexes1

Validation

+

Explain

Reset

Find

Options

```
{
  "reviews.rating": {
    "$gt": 4
  }
}
```

25

1 - 5 of 5

<

>

⋮

{}

⌂

```
member_id: 101
rating: 5
comments: "Excellent science fiction story"
1: Object (3)
  member_id: 102
  rating: 4
  comments: "Very interesting and informative"
2: Object (3)
  member_id: 103
  rating: 5
  comments: "Excellent space adventure"
```

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Cluster0

book_metadata

Cluster0 > bookflow > book_metadata

View monitoringVisualize Your Data

Documents5

Aggregations

Schema

Indexes1

Validation

+

Explain

Reset

Find

Options

```
{
  "reviews.rating": {
    "$in": [4, 5]
  }
}
```

25

1 - 5 of 5

<

>

⋮

{}

⌂

```
u: object (5)
  member_id: 101
  rating: 5
  comments: "Excellent science fiction story"
  5
1: Object (3)
  member_id: 102
  rating: 4
  comments: "Very interesting and informative"
2: Object (3)
  member_id: 103
  rating: 5
  comments: "Excellent space adventure"
```

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Cluster0

book_metadata

Cluster0 > bookflow > book_metadata

View monitoringVisualize Your Data

Documents5

Aggregations

Schema

Indexes1

Validation

+

Explain

Reset

Find

Options

```
{
  "format": {
    "$regex": "Digital",
    "$options": "i"
  }
}
```

25

1 - 4 of 4

<

>

⋮

{}

⌂

```
_id: ObjectId('6a91432f6d442b6cd5cf3e88')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
```

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Cluster0

book_metadata

+

Cluster0 > bookflow > book_metadata

Documents 5

Aggregations

Schema

Indexes 1

Validation

View monitoring

Visualize Your Data

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bookflow

book_metadata

local

studentDB

students

\$match

+

?

Explain

Run

Options

Untitled - modified

+ CREATE NEW

</>

PREVIEW

{}

</>

Stage1 \$match

ON

1 {

2 "published_year": {

3 "\$gt": 2020

4 }

5 }

Output preview after \$match stage (Sample of 4 documents)

_id: ObjectId('6a91432f6d442b6cd5cf3e88')

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

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My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
 - book_metadata
- local
- studentDB
 - students

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Cluster0book_metadata

Cluster0 > bookflow > book_metadata

Documents 5AggregationsSchemaIndexes 1Validation

\$match\$unwind

ExplainRunOptions

Untitled - modifiedCREATE NEW

PREVIEW

Stage 2 \$unwind

```
1 /**
2  * path: Path to the array field.
3  * includeArrayIndex: Optional name for
4  * preserveNullAndEmptyArrays: Optional
5  * toggle to unwind null and empty val
6  */
7 {
8   path: "$reviews"
9 }
10
```

Output preview after \$unwind stage (Sample of 10 documents)

```
_id: ObjectId('6a9143ae6d42b6cd5cf3e86')
book_id: 105
title: "Data Structures"
author: "Mark Allen"
published_year: 2024
category: "Computer Science"
format: "Digital PDF"
file_size: "5.0 MB"
member_id: 105
```

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My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
 - book_metadata
- local
- studentDB
 - students

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Cluster0book_metadata

Cluster0 > bookflow > book_metadata

Documents 5AggregationsSchemaIndexes 1Validation

\$match\$unwind\$group

ExplainRunOptions

Untitled - modifiedCREATE NEW

PREVIEW

Stage 3 \$group

```
1 {
2   "_id": "$title",
3   "average_rating": {
4     "$avg": "$reviews.rating"
5   },
6   "total_reviews": {
7     "$sum": 1
8   }
9 }
```

Output preview after \$group stage (Sample of 4 documents)

```
_id: "Data Structures"
average_rating: 4
total_reviews: 2
```

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My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
 - book_metadata
- local
- studentDB
 - students

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Cluster0book_metadata

Cluster0 > bookflow > book_metadata

Documents 5AggregationsSchemaIndexes 1Validation

\$match\$unwind\$group\$sort

ExplainRunOptions

Untitled - modifiedCREATE NEW

PREVIEW

Stage 4 \$sort

```
1 |
2 {
3   "average_rating": -1
4 }
```

Output preview after \$sort stage (Sample of 4 documents)

```
_id: "Dune"
average_rating: 4.666666666666667
total_reviews: 3
```

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My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
- book_metadata**
- local
- studentDB
- students

Cluster0

book_metadata

+

Cluster0 > bookflow > book_metadatadata

Documents 5 Aggregations Schema **Indexes 2** Validation

Create Index Refresh

Looking for search indexes?
These indexes can be created and viewed under [Search and Vector Search](#).

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR	36.9 kB	4 (since Fri Aug 28 2026)	UNIQUE	READY
> reviews.comments_text	TEXT	20.5 kB	0 (since Fri Aug 28 2026)		READY

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My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bookflow
- book_metadata**
- local
- studentDB
- students

Cluster0

book_metadata

+

Cluster0 > bookflow > book_metadatadata

Documents **5** Aggregations Schema Indexes 2 Validation

{

\$text: {

\$search: "excellent"

}

Explain

Reset

Find

Options

25 1 - 5 of 5

```
_id: ObjectId('6a91432f6d442b6cd5cf3e80')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
reviews: Array (3)
```

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Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. [Learn more](#)

Interpret

Visual Tree

Raw Output

```
{
  "stage": "FETCH",
  "inputStage": {
    "stage": "IXSCAN",
    "keyPattern": {
      "_fts": "text",
      "_ftsx": 1
    },
    "indexName": "reviews.comments_text",
    "isMultiKey": true,
    "isUnique": false,
    "isSparse": false,
    "isPartial": false,
    "indexVersion": 2,
    "textIndexVersion": 3
  },
  "parsedTextQuery": {
    "terms": ["excel"],
    "negatedTerms": [],
    "phrases": [],
    "negatedPhrases": []
  },
  "textIndexVersion": 3
}
```

Query Performance Summary

- 5 documents returned
- 5 documents examined
- 0 ms execution time
- Is not sorted in memory
- 5 index keys examined

Query used the following index:

_fts (text)
_ftsx ↑

Close